AAAI · HETEROGENEOUS GRAPH LEARNING UNDER HETEROPHILY

H2S-NN

Heterogeneous Heterophilic Sheaf Neural Network

Keeping meta-path instances as 2D-cells,
then diffusing node and edge cochains.

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ROADMAP

Outline

Following the section structure of the paper

1 **Introduction**
The joint problem, and what collapse discards

2 **Preliminaries**
Typed graphs · heterophily · schemas · sheaves

3 **Methodology**
AMPSS · Sheaf-Hodge encoding · diffusion · SheafCovPool

4 **Experiments**
Node · graph · link

5 **Discussion**
Two ablations

6 **Conclusion**
What holds, and what does not

A–F **Appendix**
Readouts · tuning · clustering · efficiency

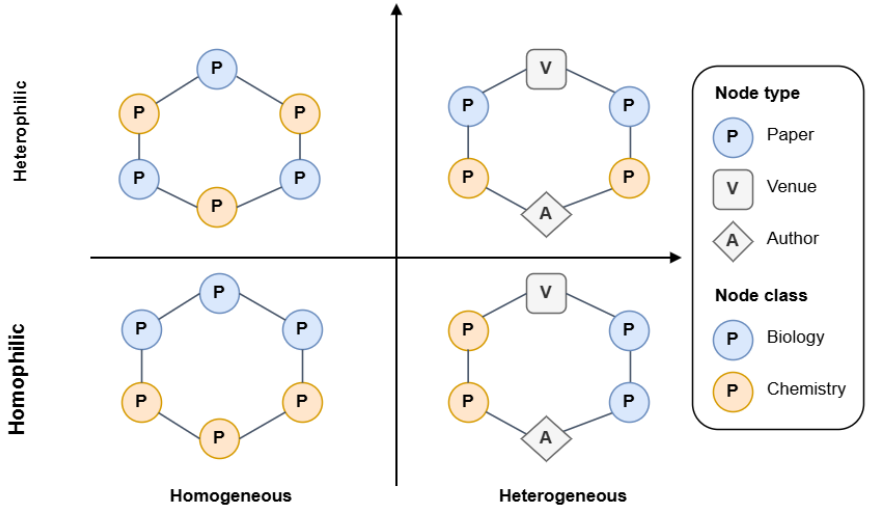
Two properties, one hard problem

- Heterogeneity — many node and relation types
- Heterophily — neighbours disagree, so averaging misleads
- Real graphs are both at once
- Assuming either away loses structure and accuracy

The literature is split down the middle

Heterogeneous methods rarely model heterophily.

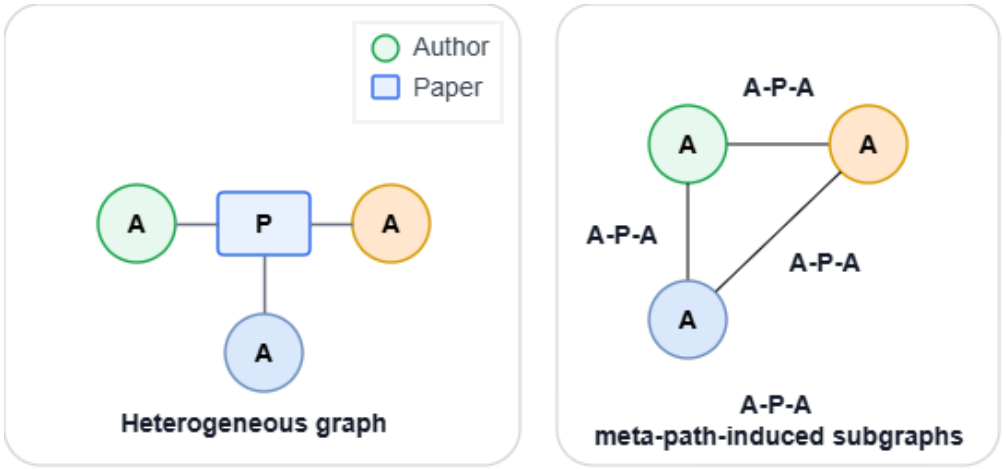
Heterophilic sheaf methods rarely model typed nodes and relations.



Several node and relation types — and adjacent nodes that disagree.

What meta-path collapse throws away

- A meta-path carries what neither endpoint holds alone
- But it is used as a view: A-P-A becomes author–author
- The shared paper leaves the propagation operator
- Under heterophily, the path decides: reinforce or contrast?



Under A-P-A, authors sharing a paper are joined directly — and the paper disappears.

A second, independent gap: granularity

Selection happens at the schema level. PAP can be useful overall while individual PAP instances link unrelated papers.

Contributions

H2S-NN

- Keep each path instance as a 2D-cell
- Build node→edge and edge→face geometry
- No homophily assumption

AMPSS

- Stage I — which schemas? (training only)
- Stage II — which instances? (learned)
- Weights act inside the upper edge Laplacian

SheafCovPool

- Transport edge states into node stalks
- Attention-weighted Gram means + covariances
- Permutation- and gauge-invariant

Evaluated on node classification, graph classification, and link prediction — where H2S-NN attains the best overall performance among the compared methods.

Typed graphs, heterophily, schema vs. instance

Heterogeneous graph

$G = (V, E, \phi, \psi)$

φ types the nodes, ψ types the edges. Heterogeneous as soon as there is more than one node type or relation type.

Homophily / heterophily

$H_{edge}(G') = (1/|E'|) \cdot \Sigma\, 1[y_u = y_v]$

Averaged over the meta-path-induced subgraphs:

$\mathcal{H} = (1/|\mathbb{P}|) \cdot \Sigma_p\, H_{edge}(G'_p)$

DBLP 59.76 · ACM 72.39 · IMDB 50.04

The distinction the paper turns on

Meta-path schema — a type-level pattern

$\pi : \; t_0 \xrightarrow{r_1} t_1 \xrightarrow{r_2} \dots \xrightarrow{r_l} t_\ell$

Induces the walk-count matrix $A_\pi = A_{r_1} \cdots A_{r_\ell}$.

Concrete path instance — one occurrence

$f = (v_0, e_1, v_1, \dots, e_\ell, v_\ell)$

Prior work selects π. We keep f.

Cellular sheaves in one slide

- A graph is a 1D cellular complex: nodes, then edges
- A sheaf gives every cell a stalk — its own local frame
- And every incidence a restriction map $F_{c \trianglelefteq \tau}$
- Diffusion runs through those maps, not over raw features

The first coboundary compares transported endpoints

$$(\delta_0\, x)_e \;=\; F_{v \trianglelefteq e}\, x_v \;-\; F_{u \trianglelefteq e}\, x_u$$

$$L_0 = \delta_0^{\top}\, \delta_0\,, \quad x^{\top} L_0\, x = \|\delta_0\, x\|^2$$

Disagreement is measured after transport, not before. Identity maps recover the ordinary graph Laplacian.

Why sheaves suit heterophily

Two neighbours can look different and still be perfectly consistent in the learned geometry.

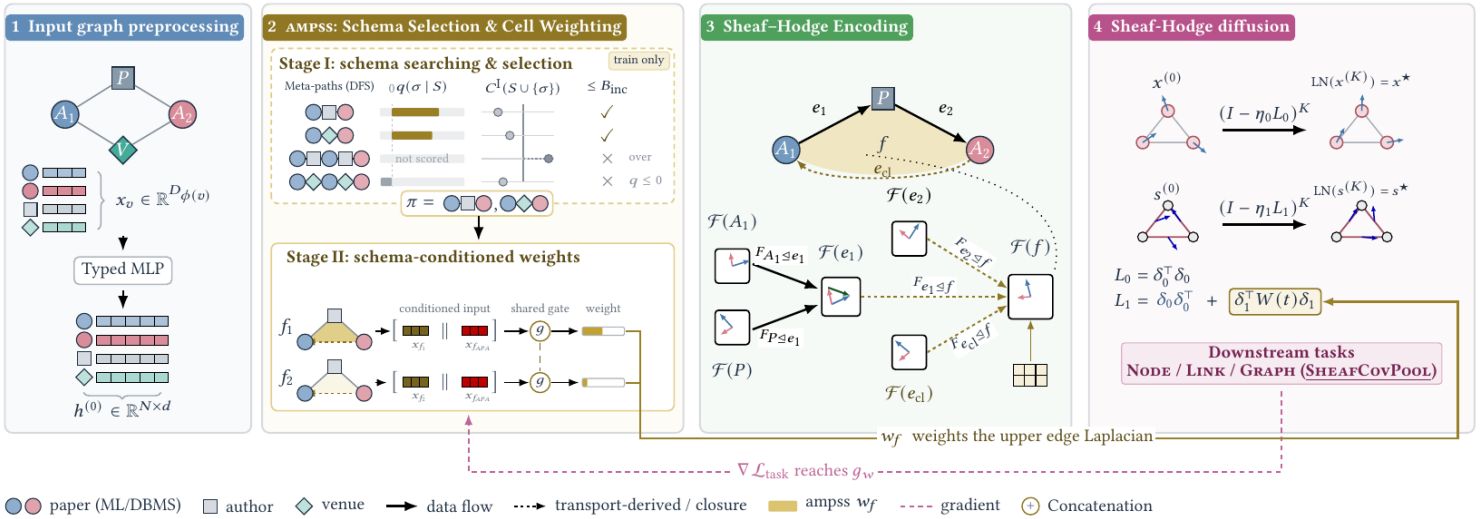
The opening we work in

Existing sheaf models — HetSheaf included — stop at observed node-edge incidences.

H2S-NN adds path-supported 2D-cells and edge-to-face maps.

Problem formulation and pipeline

$\hat{y}_o = g_\theta(o\ ;\ G,\ \mathcal{I}_\Sigma)$ one backbone; o is a node, a node pair, or a whole graph.



1 Preprocessing

Typed features → shared d-dim stalks.

2 AMPSS

Select schemas, then weight instances.

3 Encoding

Path-supported 2D-cells and their maps.

4 Diffusion

K Euler steps → task readout.

Node stalks and sheaf restrictions

Type-specific node encoder

$$h_v^{(o)} = \mathrm{Drop}[\ \mathrm{GELU}(\ \mathrm{LN}(W_{\phi(v)}\, x_v + b_{\phi(v)}) + q_{\phi(v)}\ )\ ]$$

Every type lands in one shared d-dimensional stalk space.

The edge cochain is initialised independently

$$s_e^{(o)} = \mathrm{MLP}(\ h_u \,\|\, h_v \,\|\, \zeta_r \,\|\, (\delta_0 h)_e\ )$$

Not confined to im δ_0 — an edge carries its own state.

Relation-conditioned restriction maps

$$F_{z \trianglelefteq e} = \lambda_{z,e}\, Q_{z,e}\ ,\quad Q^{\top}Q = I\ ,\quad \lambda > 0$$

The orthogonal part aligns coordinates; the scale weights the disagreement.

Why scaled-orthogonal?

Transport stays norm-preserving, so diffusion neither explodes nor collapses.

And $F^{-1} = F^{\top}/\lambda^2$ — no matrix inversion in the loop.

Everything downstream — the closure maps, the edge-to-face maps, the whole face geometry — is built from these two ingredients.

Stage I — which schemas?

Runs once, before training, on training data only

- Bounded DFS over the type graph → candidate pool Π
- Quotas fix which instances each schema retains
- Greedily add the schema with the best held-out gain
- Stop when a budget binds or nothing helps

Structural cost = edge-face incidences

$$C^{I}(S) = \Sigma_{\pi} \Sigma_{f} (\ell_f + 1)$$

A hard, auditable cap on how much structure is added.

Greedy gain on a held-out split

$$q(\pi | S) = \mathcal{L}_{sel}(S) - \mathcal{L}_{sel}(S \cup \{\pi\})$$

Subject to $|S| \le K_{\Sigma}$ and $C^{I} \le B_{inc}$.

Stated as a limitation, not a feature

Schema selection is discrete, so gradients from the task loss never reach Stage I.

For link prediction, the selection graph excludes the inner-selection positives and their reverses.

Stage II — which instances?

Fully differentiable; trained by the supervised task loss alone

A positive weight per retained instance

$$w_f = (1-r) + 2r \cdot \mathrm{sigmoid}(g_w(\xi_f \,\|\, e_\pi(f)))$$

- ξ_f — tells instances apart within a schema
- $e_\pi(f)$ — tells PAP from PSP
- r — warm-up, uniform → learned

Learning starts from the unweighted operator

The gate's output layer is zero-initialised, so $w_f = 1$ at the start.

The model departs from it only if the task loss rewards that. There is no auxiliary discovery objective.

Where the weight acts

$$\delta_1^\top W(\tau)\, \delta_1 , \qquad W = \mathrm{diag}(w_f)$$

Only the face-induced component. Since $w_f > 0$, it stays positive semidefinite.

What the weights do not touch

Closure edges stay in the extended edge domain regardless of w_f, and keep contributing to δ_0, L_0, the node-mediated term, and the readout.

Stage II modulates influence. It never prunes the complex.

Path-supported 2D-cells

- Keep the instance; do not collapse it to a shortcut
- A synthetic closure 1-cell closes the open path:

$$\partial f = (e_1, \ldots, e\ell, e_cl)$$

- It is not an observed relation, and not a shortcut
- E_ext = E ⊔ E_cl gives three graded cochain spaces
- Closure cells sit inside δ_0 , L_0 and L_1 — no side correction

The payoff, on P–A–P

The two paper-author edges stay as boundary edges of one 2D-cell.

So the intermediate author remains in the operator, instead of vanishing into a paper-paper shortcut.

Scope, stated up front

- Faces are simple paths — no revisited nodes
- Incidence signs are explicit, and are +1 here
- Enumeration is forward-only — asserted, not assumed

One free map per face — and exactness

Predict A_f, then transport it around the path

$$A_f = \mathrm{MLP}_\ell(h_{v_0} \| \cdots \| h_{v\ell} \| \zeta_{r_1} \| \cdots)$$

$$T_{v_i} = T_{v_{i-1}} F^{-1}_{u \trianglelefteq e_i} F_{v \trianglelefteq e_i}, \quad F_{e_i \trianglelefteq f} = T_{v_{i-1}} F^{-1}_{u \trianglelefteq e_i}$$

The closure maps come from the same transport — no new learned map.

Proof sketch

Consecutive path terms telescope to $T_{v\ell}x_{v\ell} - T_{v_0}x_{v_0}$, and the closure term contributes exactly its negative.

Simplicity is required because transport assigns one map per node position.

Proposition 1

$$\delta_{1,w}\, \delta_0 = 0$$

Identically — for every node cochain and every parameter value.

Enforced by construction, not by a penalty.

Why it matters

The Hodge decomposition stays interpretable: edge signals in $\mathrm{im}\, \delta_0$ are the ones nodes induce.

The face-induced term therefore acts on the residual — the part nodes cannot explain.

Sheaf-Hodge diffusion

Two operators, two lanes

$$L_0\, x = \delta_0^{T}\, \delta_0\, x$$

$$L_1\, s = \delta_0\, \delta_0^{T}\, s + \delta_1^{T}\, W\, \delta_1\, s$$

- node-mediated — through shared incident nodes
- face-induced — inside a 2D-cell; Stage II weights this

K explicit Euler steps

$$x^{(k+1)} = x^{(k)} - \alpha_0\, L_0\, x^{(k)}$$

$$s^{(k+1)} = s^{(k)} - \alpha_1\, L_1\, s^{(k)}$$

Step sizes may be learned. LayerNorm gives $x^{\star}$ and $s^{\star}$.

Matrix-free by design

Local d×d maps, batched, aggregated with indexed sums.

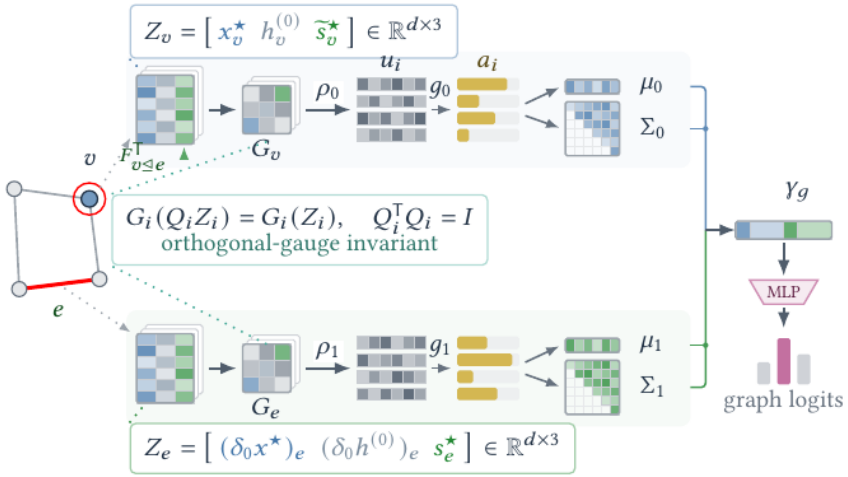
No dense Laplacian is ever materialised.

Three boundaries we state explicitly

- δ_1 uses the complete boundary cycle, not a truncation
- Signs are +1 because enumeration is forward-only
- Simple paths only — revisiting instances are excluded

SheafCovPool: a gauge-invariant readout

- Node and edge states live in different local frames
- So transport edge states into node stalks first
- Three views per cell → a 3×3 Gram matrix
- Attention → weighted mean and covariance



Two lanes: views → Gram → attention → mean and covariance.

$$\gamma_g = [\; \mu_0 \,\|\, \mathrm{triu}(\Sigma_0) \,\|\, \mu_1 \,\|\, \mathrm{triu}(\Sigma_1) \;]$$

The Gram matrix cancels the common orthogonal orientation — so γ_g is invariant to ordering AND to stalk basis.

The Gram order is fixed at 3 by the three semantic views — capacity is tuned through the post-Gram width F instead.

Settings

Task	Datasets	Protocol	Metrics
Node classification	DBLP, IMDB (HGB); ACM	HGB; transductive 20 / 20 / 60	Macro-F1, Micro-F1
Graph classification	MUTAG, PROTEINS, ENZYMES, NCI1 (TUDataset)	10-fold CV over 5 seeds	Macro-F1, Micro-F1
Link prediction	LastFM, PubMed, Amazon, YouTube (HGB)	HGB; 81 / 9 / 10; 1:1 two-hop negatives	AUPR, AUROC

Baselines, by graph type and homophily assumption

- Hom./hom. — GCN, GAT, ChebNetII
- Hom./het. — NSD, CSNN
- Het./hom. — HAN, SeHGNN, HiNormer, PSHGCN
- Het./het. — HDHGR, Hetero^{2}Net, H2SGNN, HetSheaf, LatGRL

Common protocol

- Early stopping, patience 100
- Per-dataset sweeps
- 5 seeds, mean ± std
- AMPSS Stage I sees training data only

Node classification

Macro-F1 / Micro-F1; graph homophily $\mathcal{H}$ in parentheses

Group	Model	DBLP Ma	DBLP Mi	ACM Ma	ACM Mi	IMDB Ma	IMDB Mi
Hom./hom.	GCN	90.77	90.28	91.63	91.38	62.33	67.69
	GAT	94.39	92.52	92.07	90.71	62.13	68.09
	ChebNetII	92.89	92.67	91.70	91.59	62.97	68.47
Hom./het.	NSD	92.96	93.42	92.33	92.05	61.61	65.23
	CSNN	93.73	92.12	93.13	92.81	62.92	62.59
Het./hom.	HAN	91.78	92.24	90.88	91.61	61.95	67.76
	SeHGNN	95.52	95.48	94.38	94.04	72.01	72.59
	HiNormer	94.06	94.14	91.87	89.64	64.13	67.53
	PSHGCN	95.58	95.70	93.66	94.70	71.82	74.34
Het./het.	HDHGR	95.35	95.00	93.99	94.13	59.41	59.63
	Hetero^{2}Net	93.54	93.54	94.05	92.87	65.34	70.63
	H2SGNN	95.82	95.11	94.90	94.08	74.23	75.29
	HetSheaf	93.87	93.77	95.09	94.94	64.65	67.18
	LatGRL	94.32	94.51	93.47	92.29	58.27	58.81
Ours	**H2S-NN**	**96.01**	**96.31**	**95.91**	**95.72**	**75.05**	**76.68**

Reading the table

- Best on all three, both metrics
- DBLP: +0.19 / +0.61
- ACM: +0.82 / +0.78 over HetSheaf
- IMDB: +0.82 / +1.39 — biggest gain, lowest homophily

A caution we state ourselves

Several margins sit inside one standard deviation of the runner-up.

So: best among the compared methods under this protocol — not unqualified state of the art.

Graph classification

Macro-F1 / Micro-F1; 10-fold cross-validation over 5 seeds

Group	Model	MUTAG Ma	MUTAG Mi	PROT. Ma	PROT. Mi	ENZ. Ma	ENZ. Mi	NCI1 Ma	NCI1 Mi
Hom./hom.	GCN	89.58	90.00	75.92	79.83	12.75	28.33	64.17	64.96
	GAT	89.58	90.00	75.27	78.38	14.03	16.55	64.17	64.96
Hom./het.	NSD	99.47	99.69	73.60	75.89	33.65	35.00	72.30	72.30
	CSNN	99.21	99.40	76.64	77.92	35.15	37.22	74.11	74.30
Het./hom.	HAN	82.91	85.00	79.19	82.22	46.24	45.02	77.60	77.50
	PSHGCN	94.88	95.00	70.61	72.32	60.18	58.33	74.17	74.96
Het./het.	HDHGR	84.65	85.00	77.64	81.25	58.27	58.33	79.29	79.32
	Hetero^{2}Net	89.58	90.00	74.66	78.57	61.53	61.36	63.67	65.45
	H2SGNN	94.88	95.00	70.61	72.32	62.18	61.31	64.17	64.96
	HetSheaf	100.00	100.00	80.56	82.14	62.35	63.33	79.46	79.46
	LatGRL	90.88	91.01	71.61	72.32	60.41	60.03	77.98	78.10
Ours	**H2S-NN**	**99.65**	**99.72**	**84.44**	**86.23**	**64.52**	**65.10**	**81.22**	**81.52**

What the numbers say

- Best on PROTEINS, ENZYMES, NCI1
- MUTAG: a tie, not a win — 188 graphs, saturated
- PROTEINS +3.88 over HetSheaf

The interpretation

Molecules and proteins — not the domains where heterogeneity is usually studied.

Atom and bond labels really are types, not just categorical features.

Link prediction

AUPR / AUROC; 1:1 negatives from the 2-hop neighbourhood

Group	Model	LastFM AUPR	LastFM AUROC	PubMed AUPR	PubMed AUROC	Amazon AUPR	Amazon AUROC	YouTube AUPR	YouTube AUROC
Hom./hom.	GCN	97.06	96.96	96.05	92.93	92.40	89.66	89.30	83.26
	GAT	63.06	51.14	96.07	93.27	88.62	86.10	66.76	58.40
Hom./het.	NSD	97.16	96.58	96.41	93.65	92.63	90.04	89.32	83.36
	CSNN	97.99	97.75	96.14	92.87	92.10	89.62	88.64	82.97
Het./hom.	HAN	88.26	83.32	95.78	93.62	95.24	93.64	92.46	90.10
	PSHGCN	88.02	80.55	97.58	93.27	94.28	92.34	93.15	90.66
Het./het.	HDHGR	97.89	97.93	95.03	93.15	93.37	91.42	91.68	88.33
	Hetero^{2}Net	97.58	97.45	94.66	92.57	94.71	92.34	90.67	86.45
	H2SGNN	97.88	97.22	95.22	92.32	93.18	92.31	90.17	88.96
	LatGRL	96.88	96.01	96.61	92.32	94.41	92.03	89.98	88.10
	HetSheaf	98.51	98.00	96.36	93.00	93.76	91.00	91.27	88.00
Ours	**H2S-NN**	**98.91**	**98.72**	**97.63**	**96.48**	**98.53**	**98.33**	**93.95**	**91.81**

Best on all four, all eight metrics

- LastFM, PubMed — many node types
- Amazon, YouTube — one node type
- Biggest gain on Amazon: +3.29 / +4.69

Why that is a real test

With one node type, typed endpoints tell you nothing — yet the relation sequence still carries signal.

The decoder reads pooled edge states, so the loss reaches Stage II.

Does each component earn its place?

AMPSS — DBLP node classification

Configuration	Macro-F1	Micro-F1
No path-supported 2D-cells	93.85	93.92
Predefined meta-path schemas	95.75	96.01
AMPSS, Stage I only	95.88	96.02
AMPSS, schema weight only	95.91	96.08
AMPSS, instance weight only	95.95	96.13
AMPSS (full)	**96.01**	**96.31**

- Dropping the 2D-cells costs 2.2 Macro-F1
- Hand-chosen schemas recover most of it
- Instance weighting still adds on top of Stage I

SheafCovPool — ENZYMES graph classification

Readout	Macro-F1	Micro-F1
Sum + MLP	61.25	61.79
Mean + MLP	61.42	62.09
Max + MLP	61.38	61.62
SheafPool	64.05	64.96
SheafCovPool	**64.52**	**65.10**

- Sum, mean and max all sit near 61
- Gauge invariance alone buys most of the gain
- Covariances add the rest — +0.47 / +0.14

APPENDIX A, D, E

Readouts, clustering, training dynamics

A. One backbone, three light readouts

- Node — MLP over state, encoding, pooled edges
- Link — shared projection, then dot / MLP / DistMult
- Graph — MLP(γ_g)

E. Separability during training

On a controlled four-class synthetic dataset, classes start interleaved and become compact and separated.

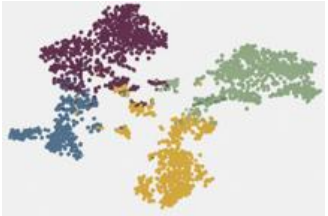
A qualitative diagnostic — not a benchmark result.

D. Node clustering (k-means, 5 seeds)

Model	DBLP NMI	DBLP ARI	ACM NMI	ACM ARI
GAT	71.50	77.26	57.22	60.47
NSD	80.21	84.04	62.56	65.02
CSNN	81.72	84.25	69.28	71.10
HAN	79.12	84.76	60.63	64.28
PSHGCN	80.38	82.65	71.89	74.31
Hetero^{2}Net	80.54	84.98	70.14	73.42
LatGRL	81.36	86.07	72.52	76.75
H2S-NN	**82.41**	**86.52**	**75.78**	**78.29**

- Best NMI and ARI on both — largest gain on ACM





APPENDIX B, F

Cost and tuning

F. Linear in graph size for fixed d and K

- Cubic in the stalk dimension d only — and $d \leq 64$
- Maps are learned only for observed edges, then cached
- Stage I runs once; SheafCovPool builds no dense matrix

B. Registered primary search space

- W&B Bayesian search with Hyperband
- $d \in \{16, 32\}$ node/link, $\{16, 32, 64\}$ graph; $K \in \{2\ldots6\}$
- Gram order fixed at 3×3; $F \in \{4, 8, 16\}$

Measured on ACM node classification

Model	Runtime (s)	Parameters	Memory (MB)
HAN	28.9	2.2M	1,780
HGT	1,025.8	7.2M	6,120
NSD	47.6	155K	980
HetSheaf	55.7	174K	—
H2S-NN	**72.1**	**517K**	**1,120**

We state the trade-off plainly

Slower and larger than HetSheaf — the price of keeping 2D-cells and diffusing both cochains.

Still an order of magnitude cheaper than HGT on every axis.

6. CONCLUSION

What holds — and what does not

Established

- Path instances can be kept as 2D-cells
- $\delta_1\delta_0 = 0$ by construction on simple-path faces
- AMPSS splits schema choice from instance choice
- SheafCovPool is permutation- and gauge-invariant

Bounded honestly

- Best among the compared methods — under this protocol
- Stage I is non-differentiable
- Proposition 1 is scoped to simple-path faces
- Costlier than the closest sheaf baseline

Best results on every evaluated node-classification, graph-classification and link-prediction dataset, with ablations supporting both AMPSS and SheafCovPool.

Next: a differentiable Stage I · faces beyond simple paths · cheaper transports, larger graphs.

Thank you

Questions welcome.

Backup material:

Proposition 1 proof · dataset statistics and semantic schemas · full search space · clustering and t-SNE · complexity derivations